

Are broker probabilities trustworthy?

A calibration audit of three production transient classifiers: ALeRCE, Fink & NEEDLE

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The decision problem

LSST: $\sim 10^7$ alerts per night. Global spectroscopy: $\sim 10^5$ per year. Brokers bridge the gap with class probabilities, which astronomers read as confidence. **That assumption has not been systematically tested against independent spectroscopic ground truth for production broker outputs.**

The calibration test:

If the broker says $P = 0.9$,
is it right **90% of the time?**

- **Overconfident** $\rightarrow$ silent failures nobody questions.
- **Underconfident** $\rightarrow$ spectra wasted on events already classified right.

Method

BTS spectra $\rightarrow$ broker $P \rightarrow$ reliability diagram $\rightarrow$
ECE $\pm$ CI $\rightarrow$ temperature scaling

$$\text{ECE} = \sum_{m=1}^{10} \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|$$

2,000-resample bootstrap 95% CIs throughout. T fitted by **held-out CV** (grouped by object) — never on the evaluation data. ECE is paired with **ROC-AUC**: an uninformative score can still look calibrated.

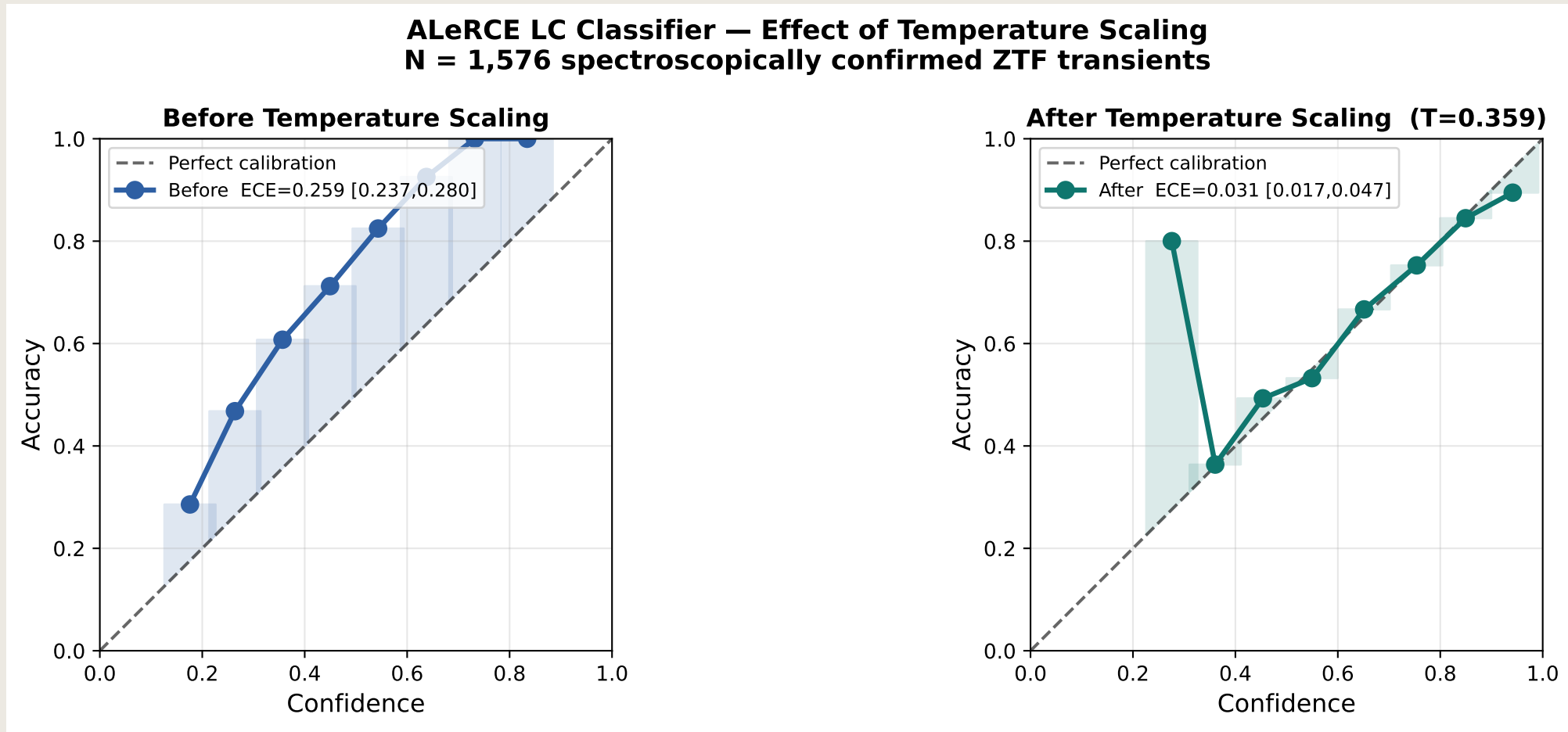
Data & key numbers

Spectroscopic labels from the **ZTF Bright Transient Survey** ($m < 18.5$) — ground truth never comes from the brokers being audited. 757 SNIa · 560 SNII · 204 SNIbc · 55 SLSN · 30 TDE. Bright, low- $z \Rightarrow$ an **upper bound** on brokers.

Classifier	raw ECE	best held-out ECE
ALeRCE LC ($n=1576$)	0.259	0.015
Fink RF ($n=1368$)	0.464	0.353 [‡]
Fink SNN ($n=1368$)	0.304	0.229 [‡]
NEEDLE-TH ($n=429$)	0.050	class-specific

[‡]best *informative* correction; isotonic gets ECE $\approx$ 0.01 only by collapsing to the base rate.

ALeRCE: underconfident — one scalar fixes the aggregate



Every bin sits **above** the diagonal: accuracy 0.71 vs. mean confidence 0.45; ECE = 0.259 [0.237, 0.280]. One temperature sharpens it. **Shown above: same-set fit**, 0.259 $\rightarrow$ 0.031 (**oracle upper bound**). **Held out: 0.015 pooled over grouped 5-fold CV** ($T = 0.361 \pm 0.011$) and **0.024 on an independent 60/40 split** — the correction generalises, with no retraining. It is not class-uniform: SNII worsens under global T (0.10 $\rightarrow$ 0.19), so class-aware calibration is the remaining step.

Confidence deficit by true class

Class	N	Acc	ECE
SNIa	757	0.83	0.377
SNIbc	204	0.83	0.362
SLSN	55	0.78	0.245*
SNII	560	0.49	0.103
TDE	30	0.00	—

*wide CI [0.18, 0.32]: suggestive only

Why? Directly observed compression:

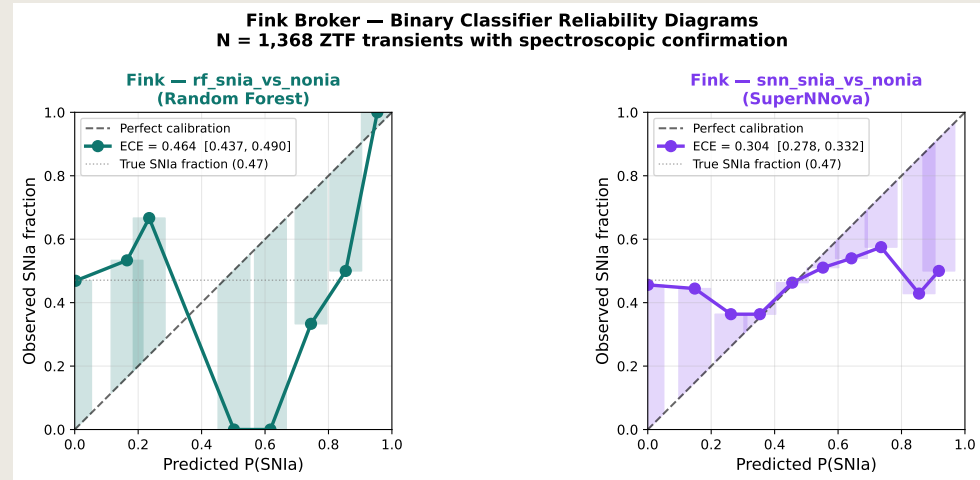
no object in 1,576 gets confidence > 0.88 ; among *correctly classified* SNIa the max is 0.72 vs. 0.83 accuracy. The RF’s vote-averaged output range cannot reach the accuracy of its best classes — deficit guaranteed.

TDE accuracy = 0.000 is not miscalibration: a headline LSST science case *absent from this classifier’s output taxonomy*.

Robustness: the result is not an artefact

- **Classifier versions.** The sample unknowingly spanned two production models. Re-queried version tags (1,601/1,606): `hierarchical_rf_1.1.0` ($n=1346$, ECE 0.258) vs. `lc_classifier_1.1.13` ($n=225$, ECE 0.260); $T = 0.38$ vs. 0.33, same sharpening regime. Pooling distorts nothing.
- **Binning.** Equal-mass (adaptive) bins move every headline ECE by $\leq 1.0\%$ — no bin-construction artefact.
- **Probability dilution** across 15 output classes explains only 3% of the miscalibration (renormalised ECE 0.251).

Fink: no ranking signal to rescale



	ECE	post- T	ROC-AUC
RF	0.464	0.353	0.502
SNN	0.304	0.229	0.525

AUC ≈ 0.5 = no ranking signal on this sample. Monotone recalibration preserves ranking, so *no* post-hoc method — temperature, Platt, isotonic, beta — can make these informative posteriors. Early-phase proxy subsets (**ndethist** ≤ 3 : AUC 0.500) don’t rescue the RF’s documented rising-SNIa regime.

NEEDLE: right on average, wrong per class

NEEDLE-TH — 5-CNN light-curve + host-image classifier (Sheng et al. 2024); **429 predictions / 278 unique objects**. Aggregate ECE = 0.050 — but a global T makes it *worse* ($\rightarrow$ 0.128): the errors are **class-asymmetric**.

Class	T_k	direction
SN	1.58 ± 0.10	overconfident
SLSN-I	1.88 ± 0.29	severely over
TDE	0.42 ± 0.04	underconfident

Take-home

- **Raw broker probabilities are not calibrated** (ECE 0.05–0.46).
- **One scalar fixes ALeRCE’s aggregate** ($T=0.36$, -94% ECE, no retraining).
- **Scalar rescaling fails for Fink**; NEEDLE needs class-specific T_k .
- **Report ECE + AUC alongside accuracy**; check taxonomy coverage.



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